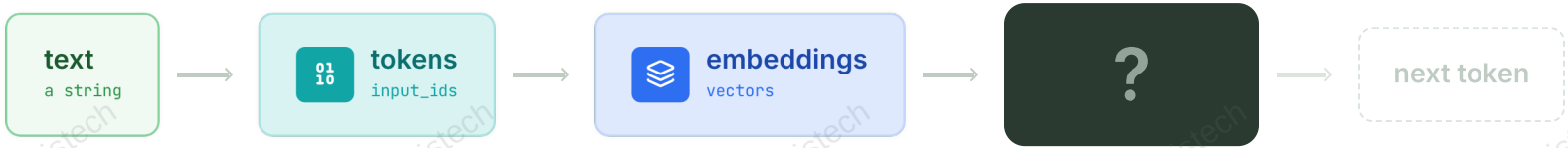
LECTURE 01

What an LLM actually eats

text → tokens → embeddings → ? → next token

Today we build the left half



IN COLOUR

Sections 5 and 6

Tokens and embeddings, in the notebook, today.

THE GREY SLAB

Stays shut

Attention and the sampling loop are lecture 2.

WHY IT MATTERS

Cost, context, bugs

Every one of them lives in the left half.

Six stops

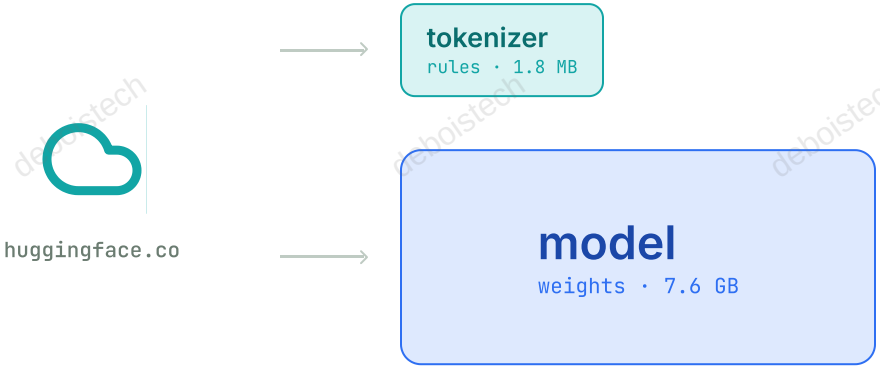
SHORT RUN	HOW WE GOT HERE	TWO PHASES	LANDSCAPE	TOKENS	EMBEDDINGS
12 min	13 min	12 min	12 min	35 min	30 min

deboistech

01 Short run

Four lines of code, three knobs, and one question you can't answer yet.

Two objects, not one



What you actually downloaded

TOKENIZER · 1.9 MB TOTAL

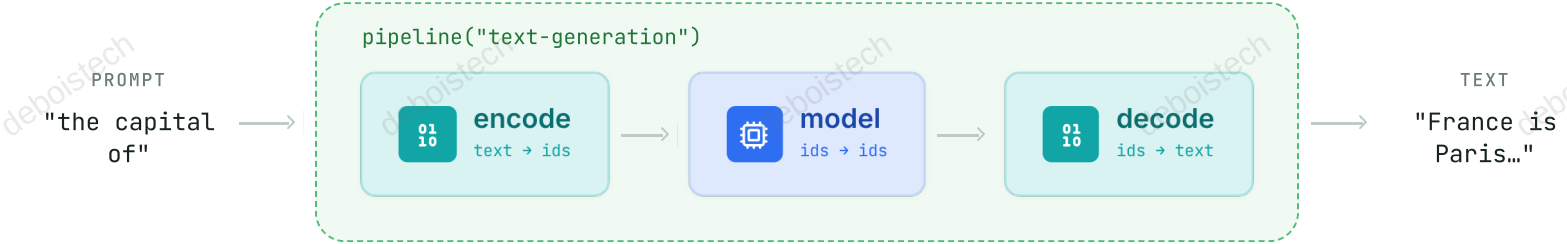
tokenizer.json	1.8 MB
tokenizer_config.json	3 KB
special_tokens_map.json	1 KB
added_tokens.json	306 B

WEIGHTS · 7.6 GB TOTAL

model-00001-of-2.safetensors	4.97 GB
model-00002-of-2.safetensors	2.67 GB
config.json	3 KB
generation_config.json	172 B

4000× the size difference. One of them decides what the other one sees.

`pipeline` is a wrapper



SCREEN

Generate your first text

01_short_run.ipynb · cell 3

What came out

PROMPT · WHAT YOU TYPED

The three laws of robotics are

GENERATED · WHAT IT ADDED

: 1. A robot may not injure a human being or, through inaction, allow a human being to come to harm. 2. A robot must obey the orders given it by human beings except where such orders would

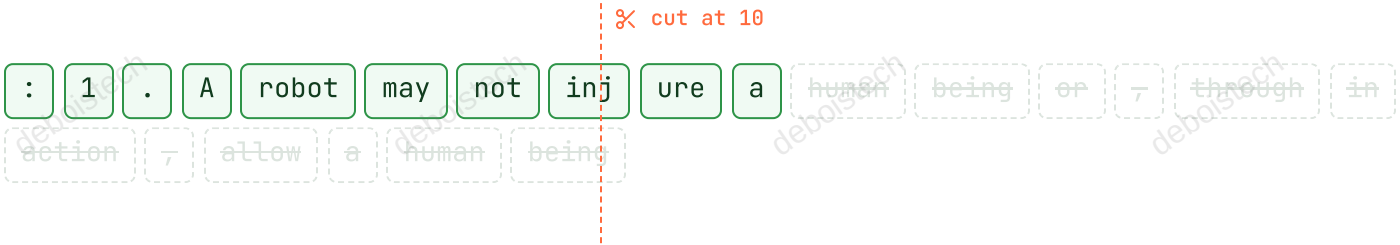
Knob 1 · max_new_tokens

: 1 . A robot may not inj ure a human being or , through in action , allow a human being to come to harm .

stop at 200

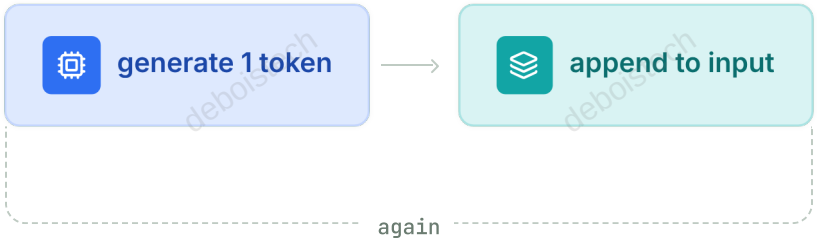
The knob does not ask for a sentence. It counts boxes and then stops.

Turn it down to 10



"...may not injure a"

It didn't finish. It ran out of turns.

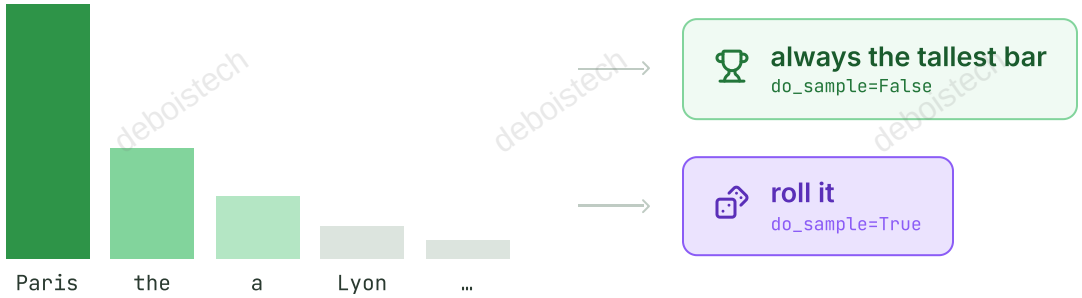


COUNTER

08 09 10

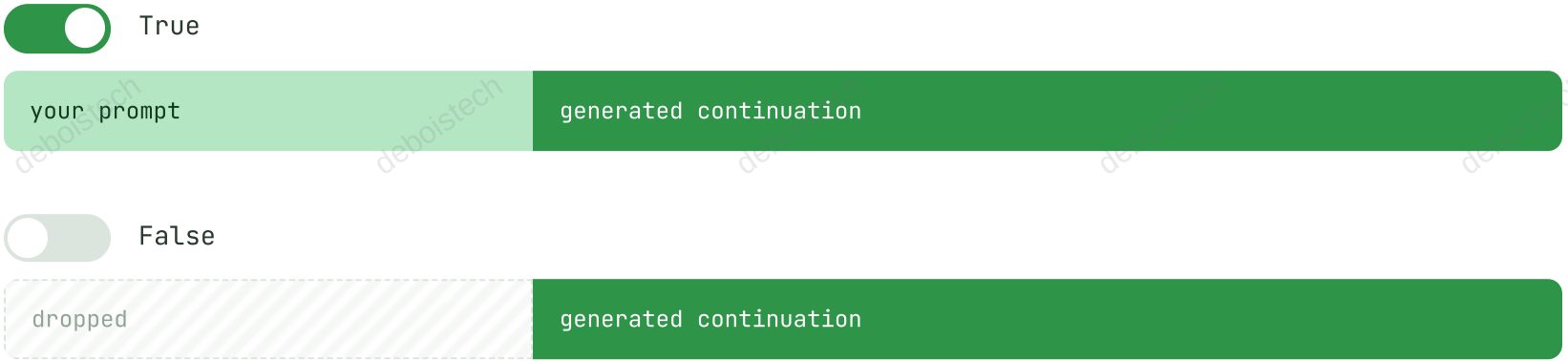
STOP

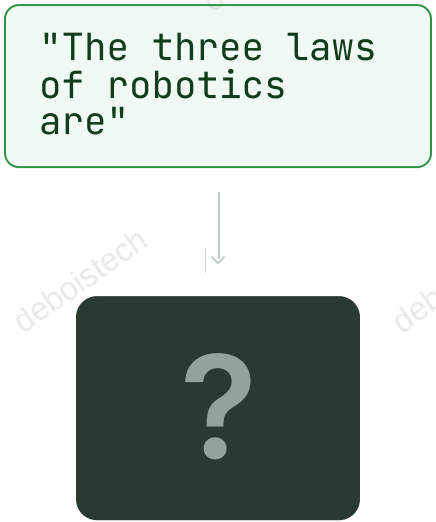
Knob 2 · do_sample



LECTURE 2 temperature, top-p and why greedy output loops

Knob 3 · return_full_text





What did the model actually receive? Not this string — *what?*

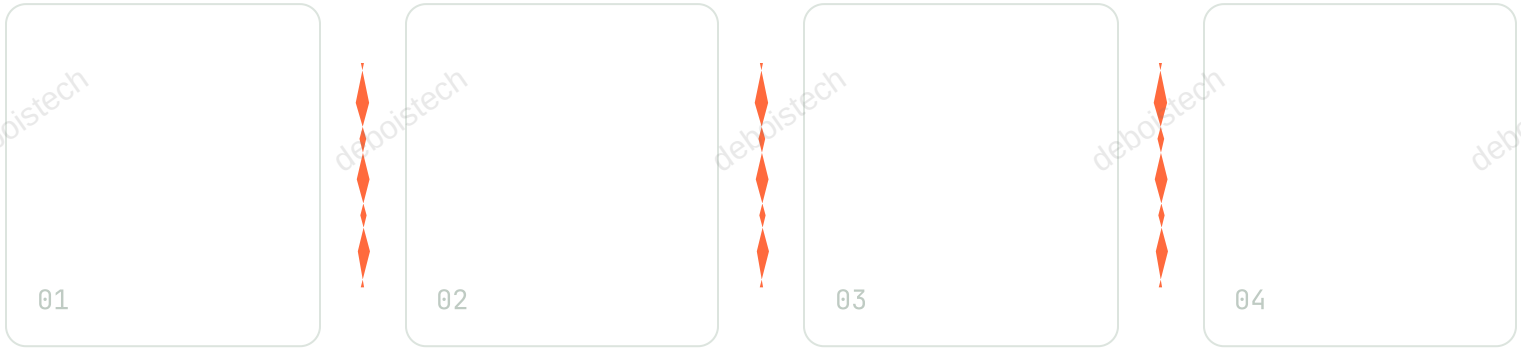
02

How we got here

Four eras. Each one is a complaint about the last one.

SHORT RUN
HOW WE GOT HERE
TWO PHASES
LANDSCAPE
TOKENS
EMBEDDINGS

Not a timeline. A chain of complaints.



Era 1 · rules and counting

GRAMMAR RULES

```
S → NP VP
  NP → Det N
    Det → the | a
    N → cat | mat
  VP → V NP
```

WORD COUNTS

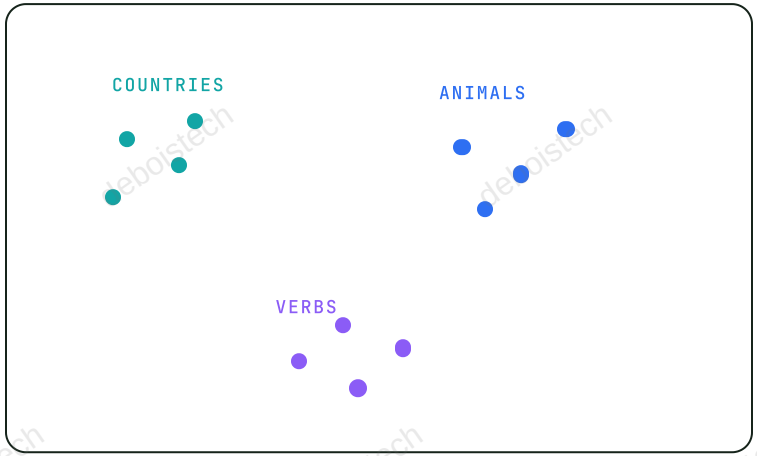
the	18,229
of	9,411
cat	1,204
defenestrate	2

Language is all exceptions

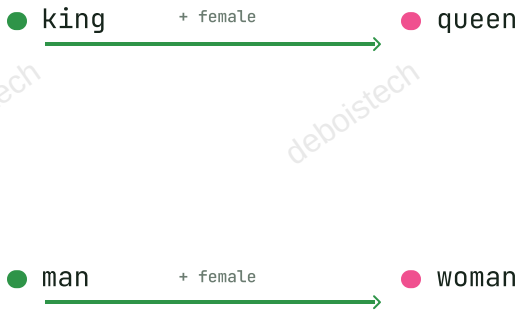
N → cat | mat

> …except "fish" (plural = fish)
>
> …except "fishes" when species
>
> …except in idiom "other fish to fry"
>
> …except sarcasm inverts it entirely
>
> …except "time flies like an arrow"
>
> …except "fruit flies like a banana"
>
> …except new slang, coined this morning
>
> …except typos, which mean the same thing
>
> …except names, which follow no rule at all
>
> …except code, which is a different language

Era 2 · words become vectors



Meaning as geometry



THE CLAIM

The same direction means the same thing twice

One word, one vector, forever

…sat on the river **bank**

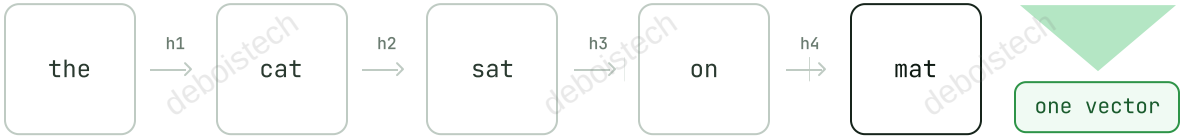
…withdrew cash from the **bank**



bank

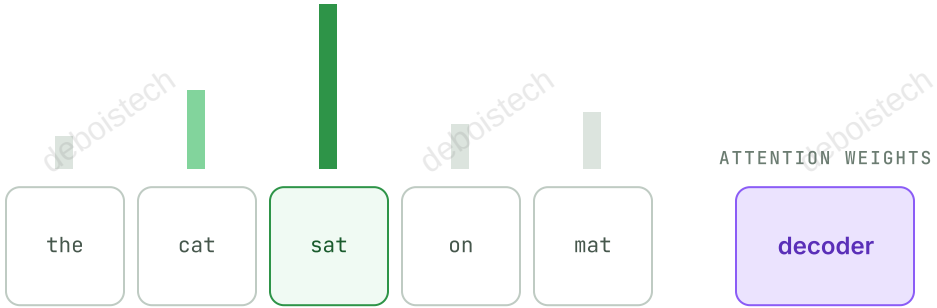
One row in the table. No context can move it.

Era 3 · read it in order



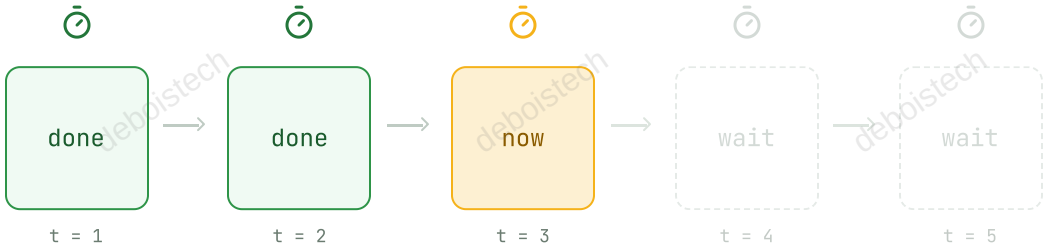
Everything the sentence meant has to fit in the last state.

Bolt on attention



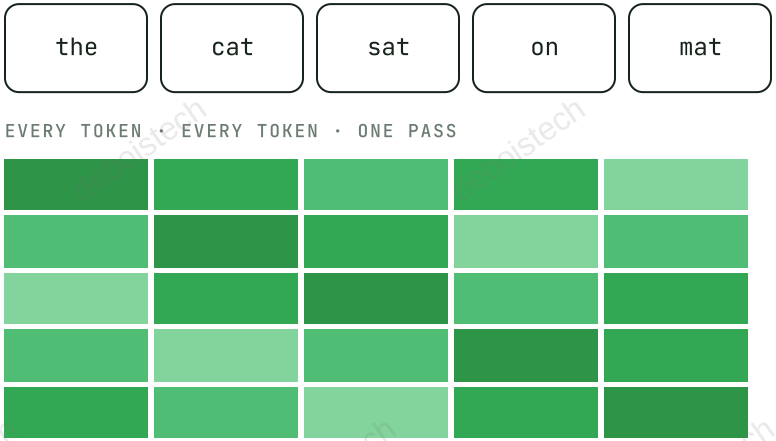
It can look anywhere. It still has to wait for everything.

Everything waits its turn



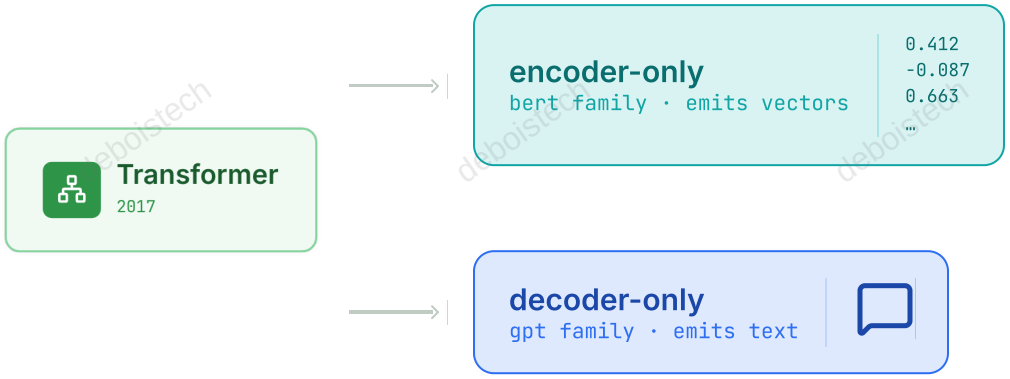
A queue. Adding GPUs does not shorten a queue.

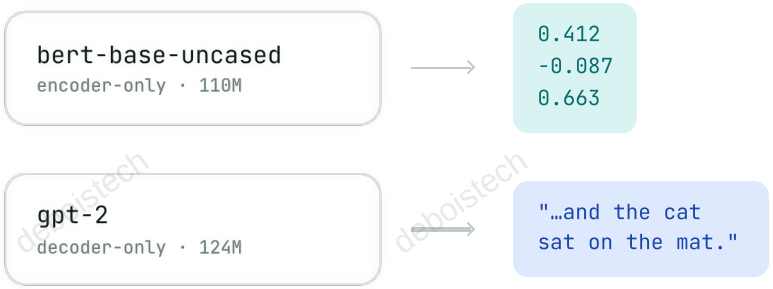
Era 4 · drop the recurrence



one clock
for the row

The fork


Transformer
2017
encoder-only
bert family · emits vectors
0.412
-0.087
0.663
…
decoder-only
gpt family · emits text



Vectors, or text?

03 Two phases

One enormous phase teaches it language. A tiny one teaches it manners.

Why does prompting work at all?



Nobody wrote a rule for "summarise". So where did the obedience come from?

Two phases, to scale

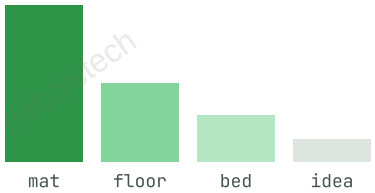


~99% OF THE COMPUTE

Fine-tuning

thousands of examples · days

the cat sat on the ?



predict the next token.

Everything it could reach



No labels anywhere

CLASSIC SUPERVISED LEARNING

X	Y · HUMAN-PROVIDED
"loved every minute"	positive
"walked out at 20 min"	negative
"it was fine, I guess"	neutral

Somebody was paid per row. That is your ceiling.

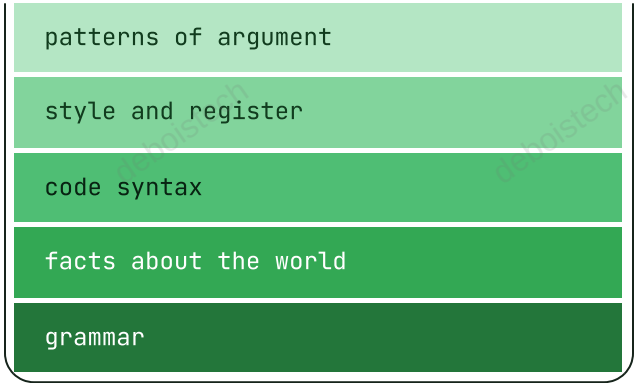
PRETRAINING

the cat sat on the **mat**

FREE The label is the next word in the same sentence.

Every sentence ever written is already labelled. That is why this scaled and the left column didn't.

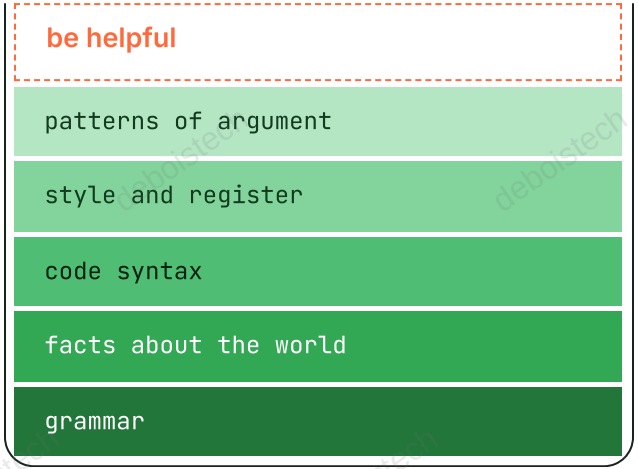
What it absorbs on the way



SEDIMENT

All of it is a side effect of one guessing game

What it does not get



MISSING

Nothing in the game rewards answering you

SCREEN

Ask a base model a question

03_two_phases.ipynb · cell 2

It continued the document

What is the capital of France?

What is the capital of Germany?
What is the largest city in Europe?
What is the population of Paris?
What is…

A plausible continuation. Not an answer.

Phase 2 · fine-tuning

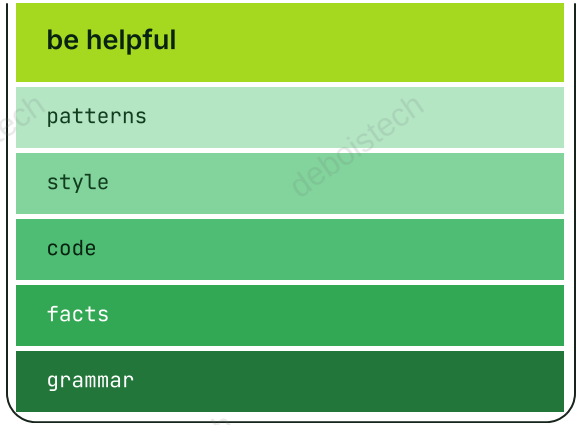
INSTRUCTION → GOOD RESPONSE

"Summarise this email."
→ Two clean lines.

"What is the capital of France?"
→ Paris.

"Fix this function."
→ The patch, then why.

≈ 10k–100k pairs



The suffix is not decoration

`Phi-3-mini` → `"What is the capital of France? What is the capital of Spain? What is…"`

`Phi-3-mini-instruct` → `"Paris."`

Same knowledge. One of them has been taught what a question is for.

Where you live



Inference

prompting rag tools agents



train a model

1000 GPUs · 3 months



use one well

a laptop and an API key

04

Landscape

Somebody else's GPU, or yours. The answer is usually both.



Closed API

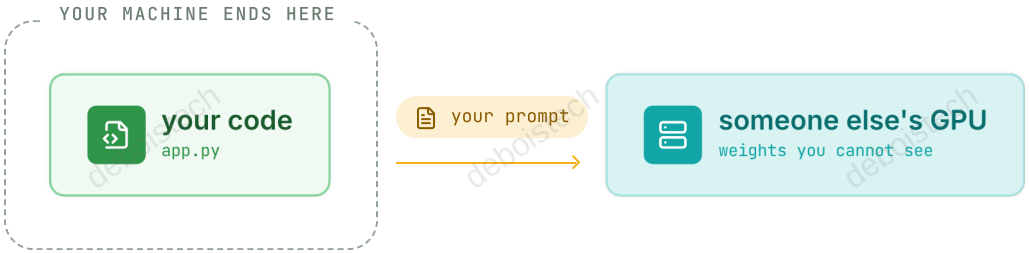
You send text. Someone else owns the weights.



Open weights

You download the file. It runs on your hardware.

Closed API



YOU GET

the strongest models, no ops, per-token pricing

YOU GIVE UP

the data, the version, the uptime

WATCH FOR

the model changing under you on a Tuesday

Open weights



YOU GET

privacy, a frozen version, zero marginal cost

YOU TAKE ON

VRAM limits, ops, a weaker ceiling

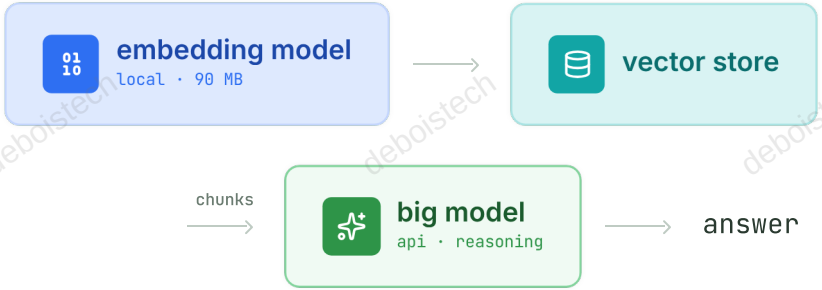
READ

the licence — "open weights" is not "do anything"

Four sliders



You'll use both



OWNED

Retrieval runs millions of times. It has to be free and private.

RENTED

Reasoning runs once per question. Buy the best one.

VRAM is the real constraint



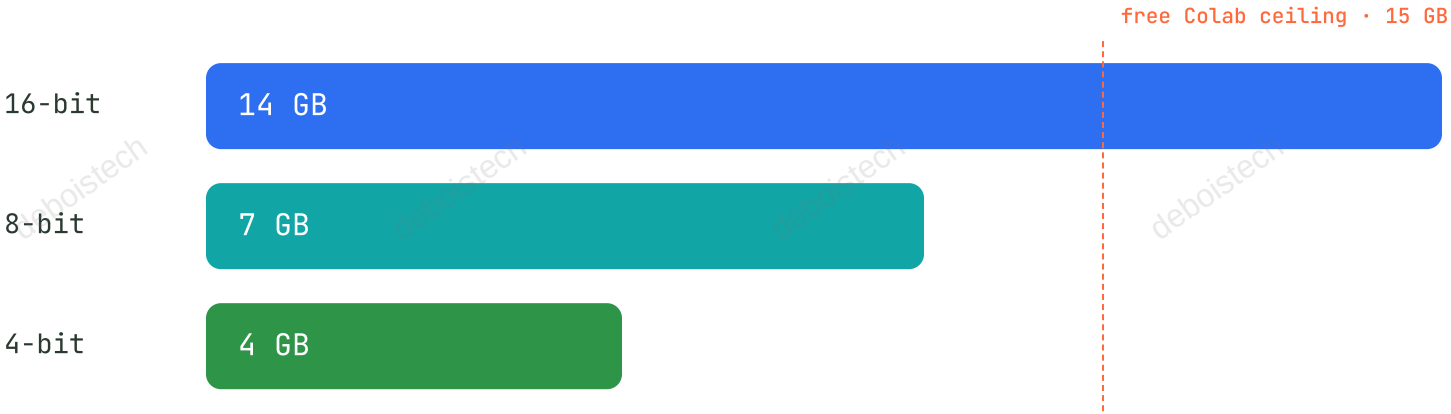
If it doesn't fit, it doesn't run. There is no partial credit.

The rule of thumb



Then add roughly 20% for the KV cache and the rest of the runtime.

Quantization shrinks the bar



You lose a little accuracy. You gain the ability to run it at all.

What fits where

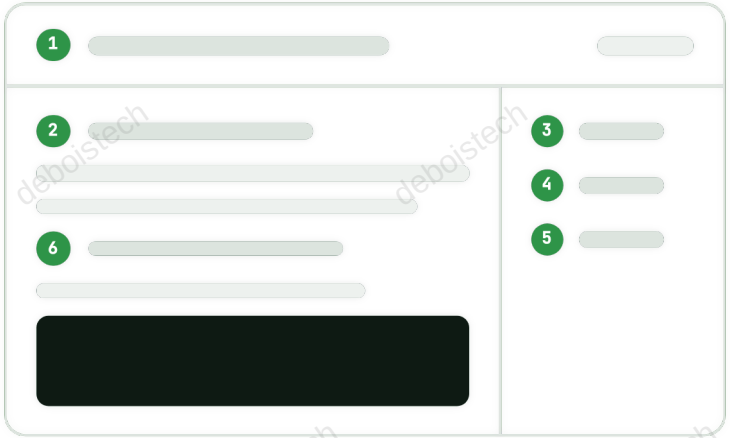
COLAB FREE TIER	embedding models	phi-3-mini 4-bit	7B 4-bit
ONE CONSUMER GPU	7B 16-bit	13B 8-bit	34B 4-bit
API ONLY	frontier models	400B+	

SCREEN

A real model card

huggingface.co/microsoft/Phi-3-mini-4k-instruct

Model card anatomy



01 parameter count

02 base vs instruct

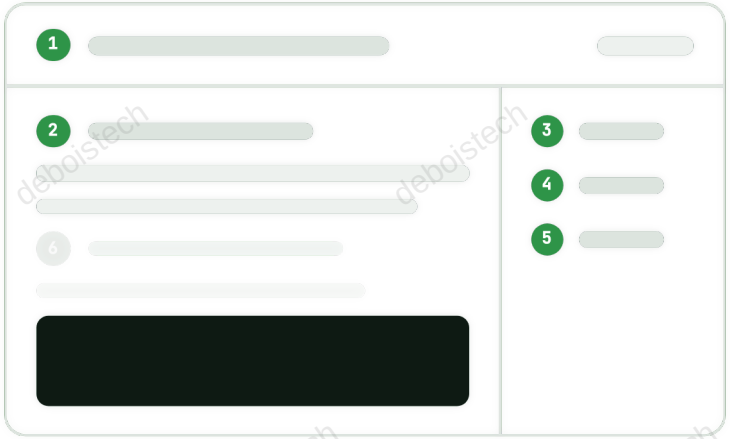
03 context length

04 licence

05 tokenizer

06 intended use

The five you actually read



01 **parameter count** — will it fit?

02 **base vs instruct** — will it answer?

03 **context length** — how much fits in?

04 **licence** — can you ship it?

05 **tokenizer** — what will it cost?

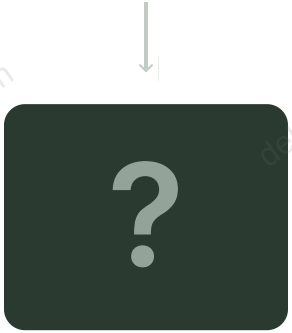
06 ~~intended use~~

Tokens

05

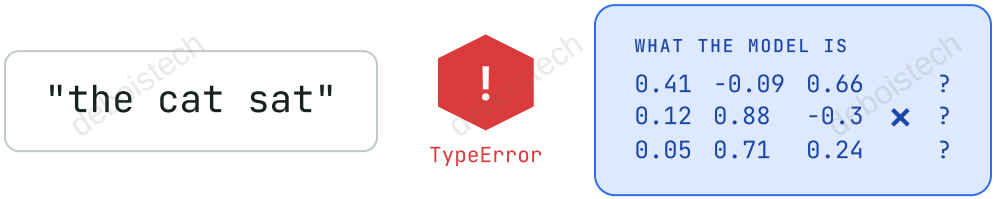
The answer to the question you have been holding for an hour.

"The three laws of robotics are"



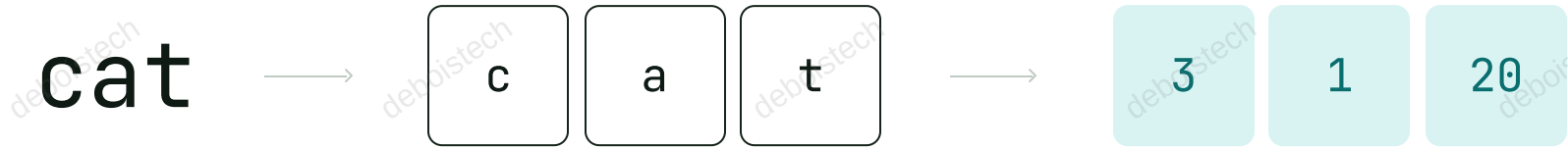
What did the model actually receive?

The mismatch



Every layer is a multiply. So the first job of the stack is to turn text into numbers.

One number per character



Vocabulary of about 100 symbols. Nothing ever falls outside it.

t

MEANING CARRIED

One letter tells the model almost nothing

"THE CAT SAT ON THE MAT" · 22 BOXES

t	h	e		c	a	t		s	a	t		o	n		t	h	e		m	a	t

Six words. Twenty-two positions to pay for.

One number per word

VOCABULARY	
the	1
cat	2
sat	3
on	4
mat	5



Six words, six positions. Each one carries real meaning.

The vocabulary never closes

the	1
cat	2
…	…
zygote	50,000
rizz	?
recieve	?
Adarsh	?
gpt-4o-mini	?

`rizz` `recieve` `Adarsh` `gpt-4o-mini` `deboistech`



[UNK]

one index for everything it has never seen

And it wastes what it has



Four separate rows to learn. The shared "run" is thrown away.

The oldest answer

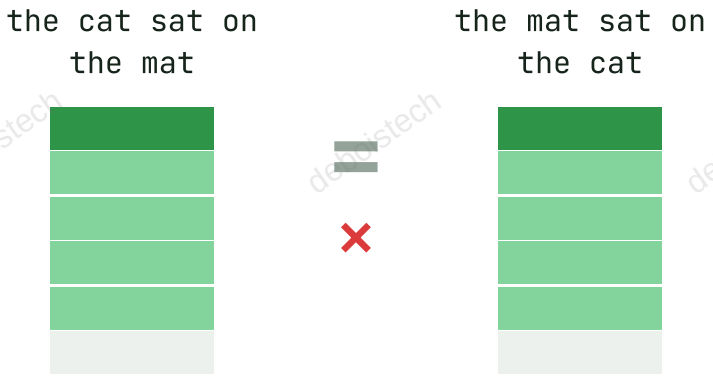
```
the cat sat on
the mat
```


the	2
cat	1
sat	1
on	1
mat	1
dog	0
ran	0
…	0

ONE CELL PER VOCABULARY WORD

50,000 cells long. Six of them are non-zero.

It throws away the order



Identical vectors. One of these sentences is about a cat.

Attempt 3 · subword

unbelievable

un | believ | able

rare · splits into pieces

the

the

common · stays whole

Nothing is ever unknown, because the pieces always exist.

Why subword won

	SMALL VOCABULARY	SHORT SEQUENCES	HANDLES NEW WORDS
character	●●●	○○○	●●●
word	○○○	●●●	○○○
subword	●●○	●●○	●●●

SCREEN

input_ids

05_tokens.ipynb · cell 1

"The three laws of robotics are"

791	2380	6634	315	6957	1233	527

"The three laws of robotics are"



This is the entire view.

SCREEN

Decode each ID, one at a time

05_tokens.ipynb · cell 2

The seams are in the wrong places

Token	isation	is	un	int	uit	ive	at	first	.

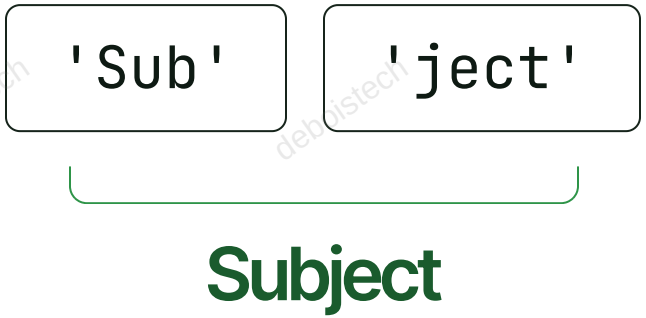
WHAT YOU SEE

Seven words, six spaces, one full stop.

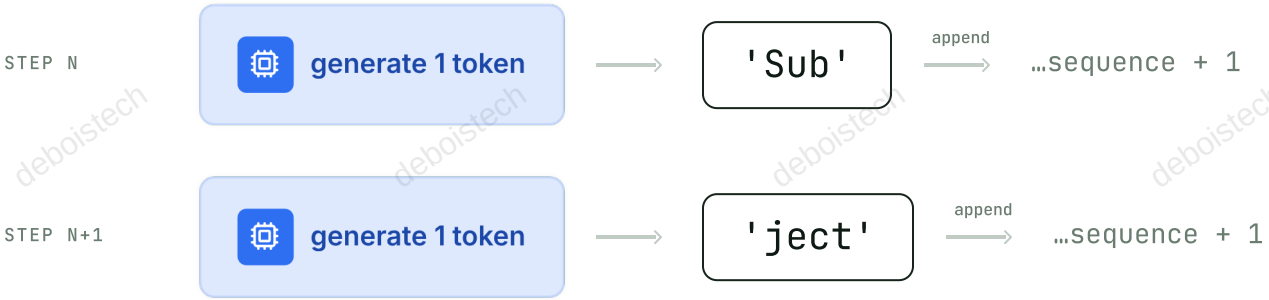
WHAT IT SEES

Ten fragments, and the space belongs to the token after it.

Words arrive in pieces



It never chose "Subject"



Two decisions, made a moment apart. The word is an artefact of printing.

Three families

MERGE THE FREQUENT PAIR

BPE

gpt-2 · gpt-4
llama
mistral

LIKELIEST SPLIT

WordPiece

bert
distilbert
electra

NO SPACES NEEDED

SentencePiece

t5
gemma
multilingual models

bert-base-uncased	
model_type	bert
tokenizer_class	WordPiece
vocab_size	30,522

gpt-2	
model_type	gpt2
tokenizer_class	BPE
vocab_size	50,257

gemma-2-9b	
model_type	gemma2
tokenizer_class	SentencePiece
vocab_size	256,000

The test string

```
CAPITALIZATION is loud 🤯 नमस्ते दुनिया ····return
12.0*50=600
```

- **CAPITALS** — some tokenizers lowercase everything
- **EMOJI** — four bytes, not one character
- **NON-ENGLISH** — trained mostly on English text
- **INDENTATION** — code models learn whole indents
- **NUMBERS** — digits group by frequency, not value

Predict first

BERT	GPT-2	GPT-4	code model	multilingual
?	?	?	?	?

SCREEN

Tokenizer playground

five tokenizers · one string

BERT

31 tokens

capital ##ization is loud [UNK] न ##म ##स ##ते द ##नि ##या return 12 .
0 * 50 = 600

Whitespace is gone entirely. The emoji became one unknown.

The capitals are gone

BEFORE

`CAPITALIZATION`

AFTER

`capital` `##ization`

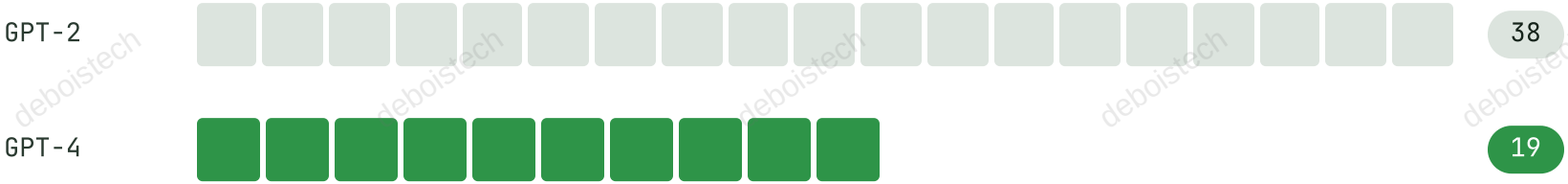
Lowercased, then shredded. "uncased" is not a detail.

GPT-2

38 tokens

| | | |

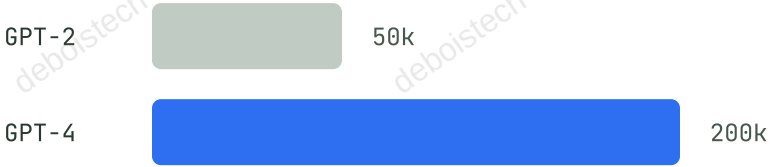
GPT-4, same string



Half the tokens for identical text. Same prompt, half the bill.

Bigger vocabulary, fewer tokens

VOCABULARY SIZE



TOKENS FOR THE SAME TEXT



More rows in the lookup table, fewer positions per sentence. That is the trade.

Code model vs indentation

CODE MODEL



2 tokens

TEXT MODEL



5 tokens

Same line of Python. Use the code tokenizer for code.

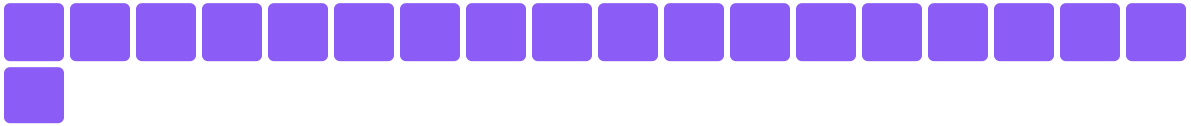
Non-English explodes

"Hello world, how are you?"



7 tokens

"नमस्ते दुनिया, आप कैसे हैं?"



19 tokens

×2.7 same sentence, same meaning, nearly three times the tokens

The emoji shatters





`\xf0` `\x9f` `\xa4` `\xaf`

Four tokens, individually unreadable, reassembled only on your screen.

You are billed per box

	TOKENS	PRICE / 1M	1M REQUESTS
GPT-2 tokenizer	38	$2.50	**$95**
GPT-4 tokenizer	19	$2.50	**$48**

Identical text, identical price per token, half the invoice.

The non-English tax

	TOKENS	PRICE / 1M	1M REQUESTS
English	7	$2.50	$18
Hindi	19 ×2.7	$2.50	$48

Same product, same question, 2.7× the running cost per user.

Context is measured in these

4,096 TOKENS

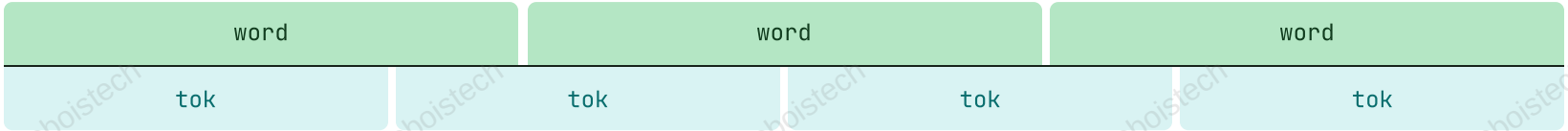
The ·quar terly ·report ·shows ·rev enue ·up ·18 % ·against ·a ·soft ·compar ison ·quarter …

NOT WORDS · NOT CHARACTERS

Some of the boxes are half a word

So you cannot count your document in words and expect it to fit.

A rule of thumb

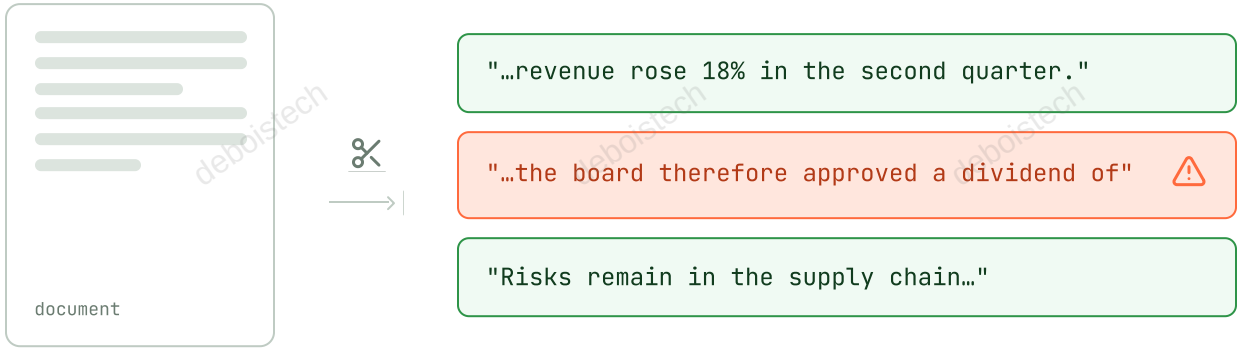


one page of prose
≈ 500 words

→

≈ 670 tokens
so a 4k window ≈ 6 pages

Chunking cuts where you tell it to



The middle chunk retrieves perfectly and answers nothing.

"How many r's in strawberry?"



3 opaque boxes · not 10 letters

bounces off the outside

It is not bad at counting. It cannot see the letters at all.

The digits don't line up

PLACE VALUE	10^6	10^5	10^4	10^3	10^2	10^1	10^0
THE NUMBER	3	7	2	1	8	4	5
ITS TOKENS	372			18			45

Three tokens, no alignment with columns. Give it a calculator instead.

A trailing space is a different prompt

"the capital of France is"

1820 6864 315 6081 374

"the capital of France is·"

1820 6864 315 6081 374 220

One extra id, and now the next token has to start a word after a space.

- cost →
- context →
- letter counting →
- arithmetic →
- whitespace →

It only ever sees tokens.

06 Embeddings

Where similar starts to mean close.

WHERE THIS IS GOING

Everything you build in week 3 needs this

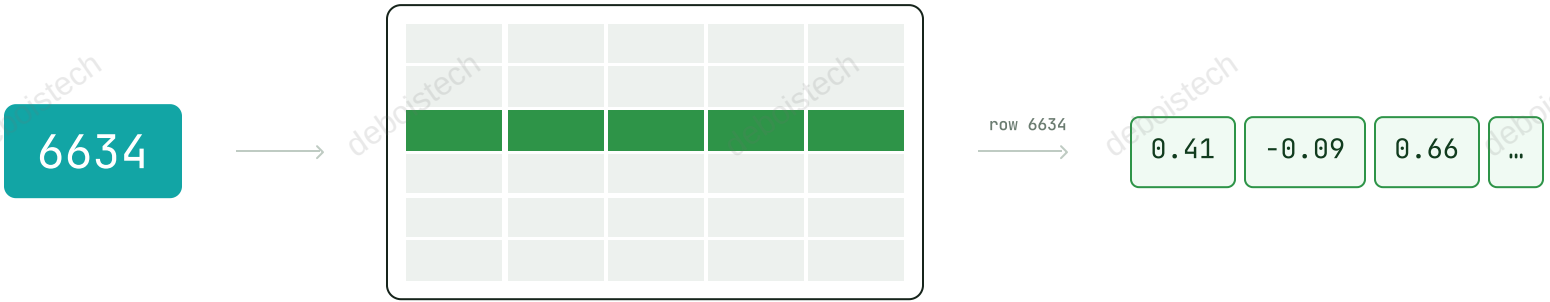


Integers carry no meaning



The id is a row number in a dictionary. Nothing more.

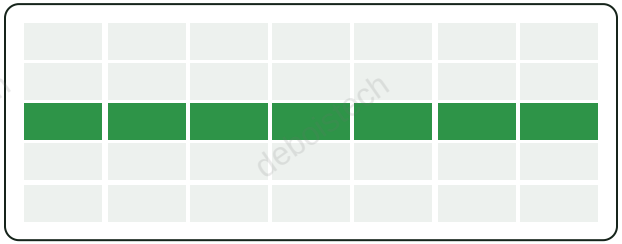
The lookup table



The shape

vocabulary size

50,257



hidden dimension

768

TOTAL

38.6M

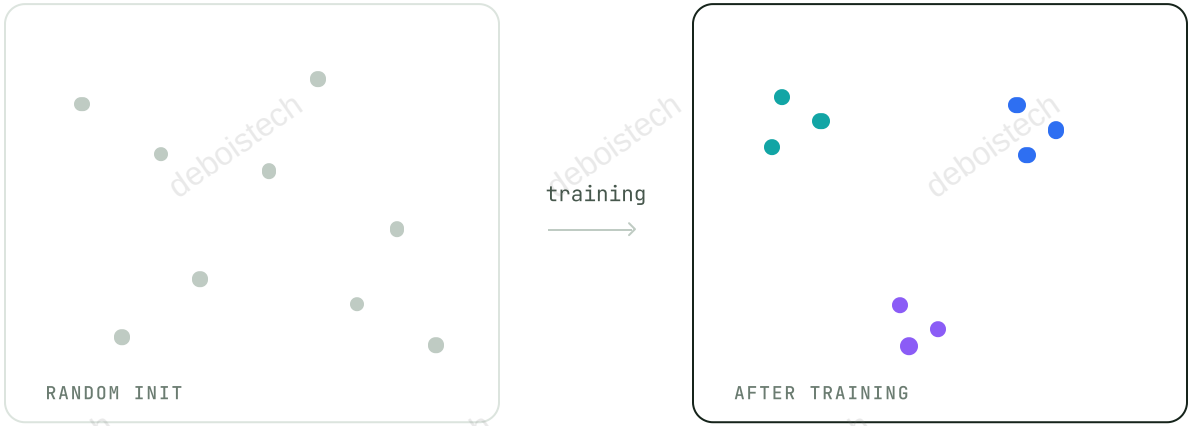
numbers, before any layer runs

Part of the model is just data

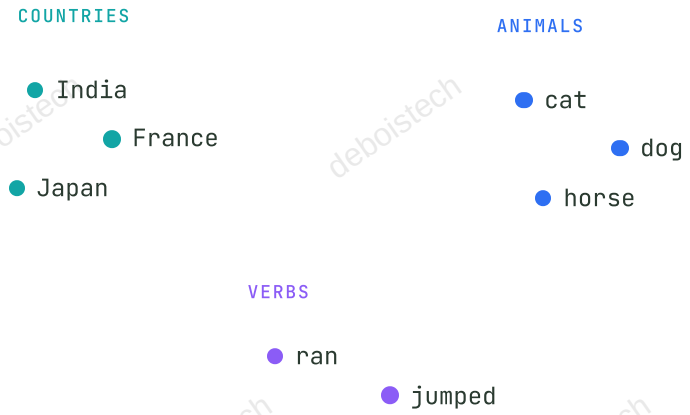


The first thing your prompt touches is a table you could open in a spreadsheet.

Where do the numbers come from?



Similar means close

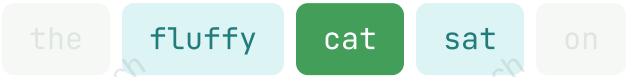


NOBODY LABELLED THESE

The clusters fall out of usage alone

Learned from neighbours



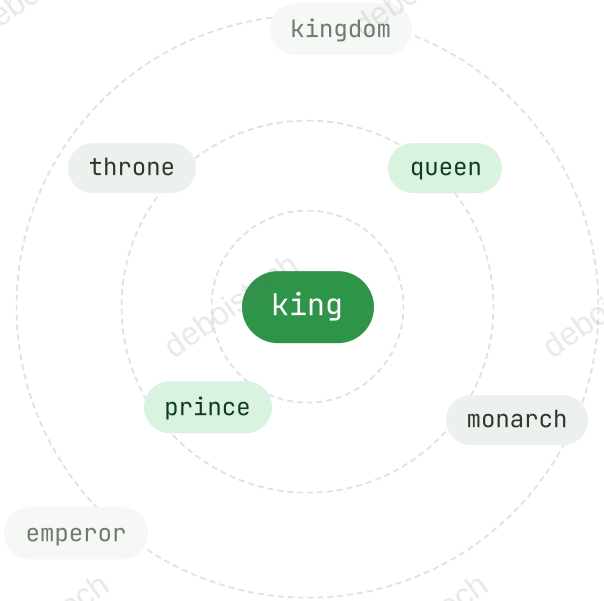


predict a word's neighbours.

SCREEN

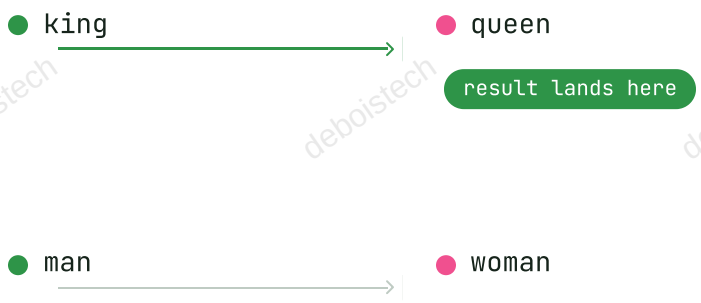
Nearest neighbours

06_embeddings.ipynb · cell 1



queen	0.74
prince	0.71
monarch	0.66
throne	0.61
kingdom	0.55

king – man + woman ≈ queen



THE SAME ARROW, TWICE

Direction carries the relationship

One word, one fixed point



bank

WORD2VEC · GLOVE

Looked up before the sentence is read

Two meanings, one dot

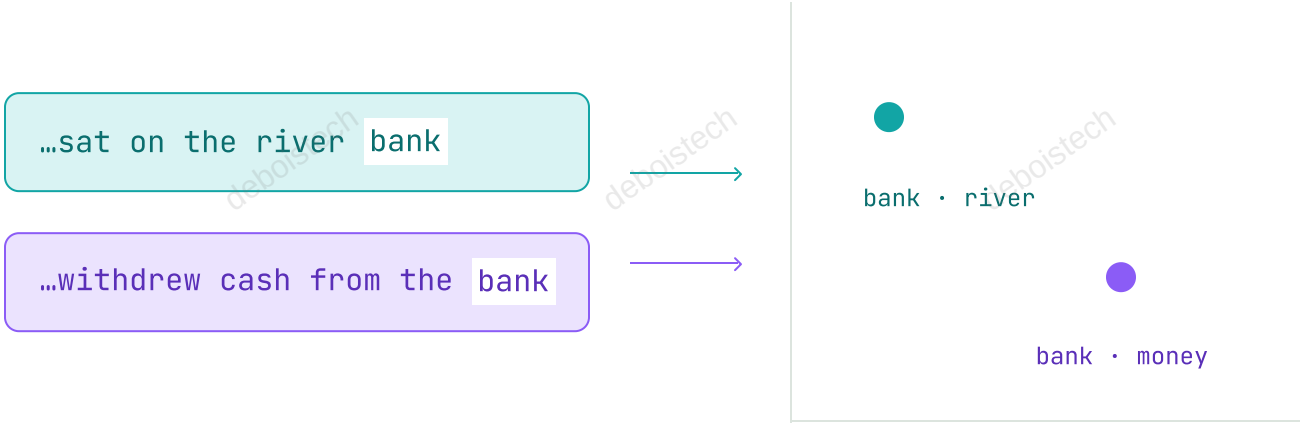
…sat on the river `bank`

…withdrew cash from the `bank`



bank

Let context move the vector



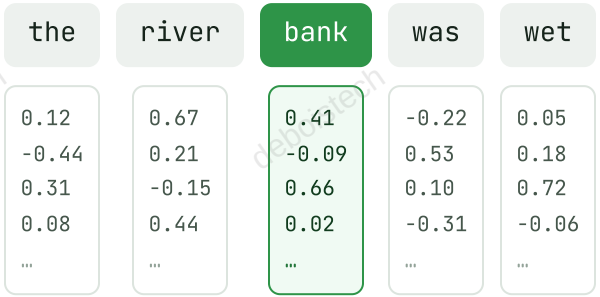
Same token id. Two different vectors, because the neighbours differ.

SCREEN

Contextual embeddings

06_embeddings.ipynb · cell 4

One vector per token, in context



Five tokens in, five vectors out. Each one 768 numbers long.

Same word, different vectors

"the river bank was wet"

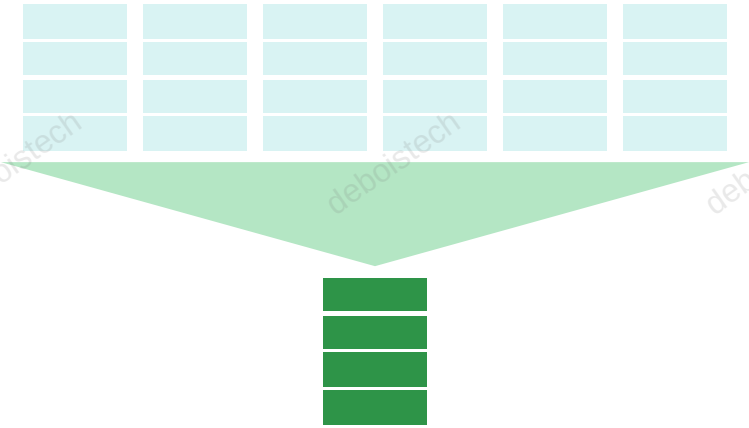
```
0.41
-0.09
0.66
0.02
```

"the bank raised rates"

```
-0.18
0.72
0.11
-0.35
```



From many vectors to one



one vector for the whole passage

Mean pooling

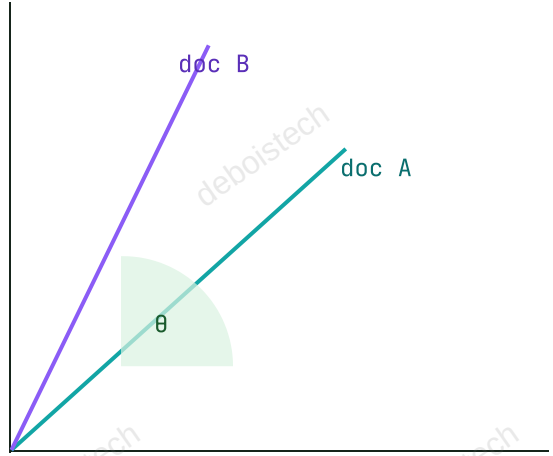
the	0.12	-0.44	0.31	0.08
river	0.67	0.21	-0.15	0.44
bank	0.41	-0.09	0.66	0.02
MEAN				
passage	0.40	-0.11	0.27	0.18

A text embedding model



One call in, 384 numbers out. 90 MB on disk, runs on a laptop.

Measuring closeness



COSINE OF THE ANGLE



-1 opposite 0 1 identical

Length is ignored. Only direction counts, so a long document and a short question are comparable.

The test set

S1

Q3 revenue rose sharply.

S2

Sales grew a lot last quarter.

S3

Q3 revenue forms are printed on recycled paper.

S4

Sales grew tomatoes on the office roof.

s1 and s2 mean the same thing and share no words. s3 and s4 share the words and mean something else.

What word-matching would say

	s1	s2	s3	s4
s1	1.00	0.04	0.71	0.02
s2	0.04	1.00	0.06	0.68
s3	0.71	0.06	1.00	0.09
s4	0.02	0.68	0.09	1.00

THE WRONG PAIRS

Confident, and backwards

SCREEN

Cosine similarity matrix

06_embeddings.ipynb · cell 6

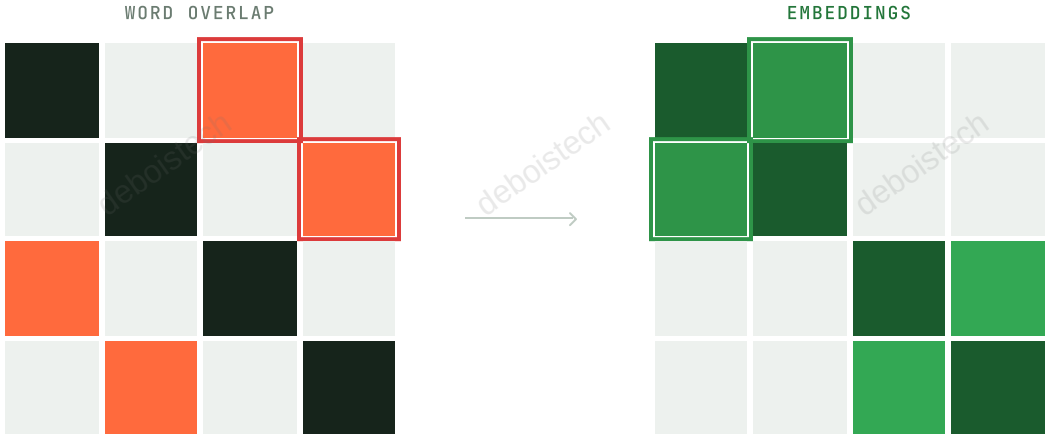
What embeddings say

	s1	s2	s3	s4
s1	1.00	0.79	0.31	0.14
s2	0.79	1.00	0.18	0.27
s3	0.31	0.18	1.00	0.62
s4	0.14	0.27	0.62	1.00

THE RIGHT PAIRS

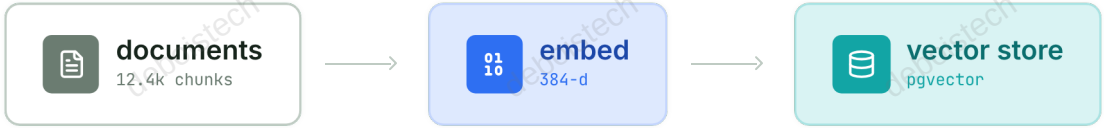
s1 with s2. s3 with s4.

Meaning beat word-matching

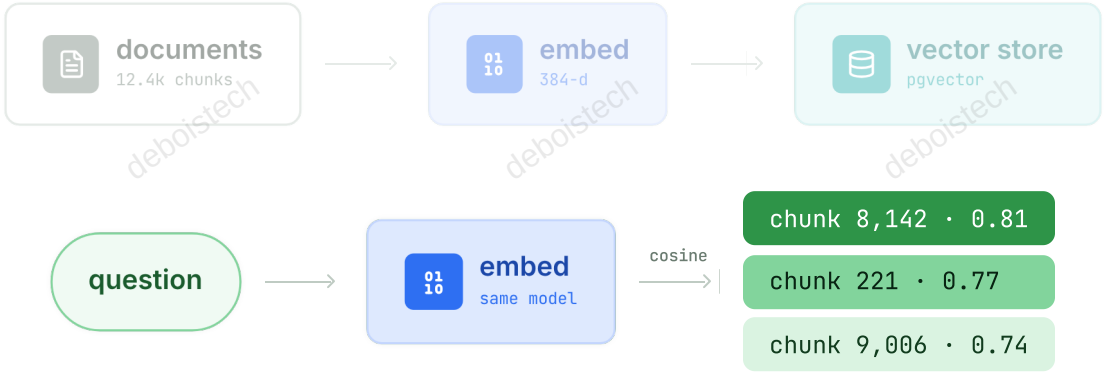


The high cells moved to the correct pairs. Nothing about the sentences changed.

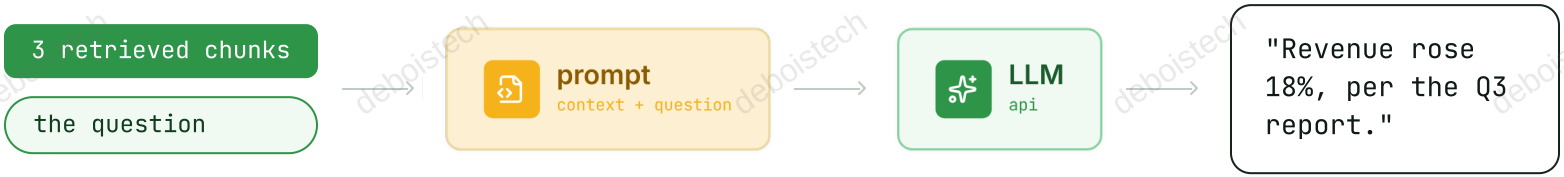
Index once

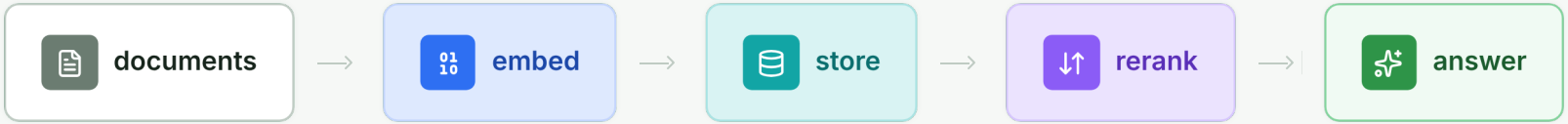

documents
12.4k chunks
01
10
embed
384-d
vector store
pgvector

Ask in the same space



Retrieved text goes in the prompt





WEEK 3 chunking, reranking, and why your recall is bad

It works on any sequence

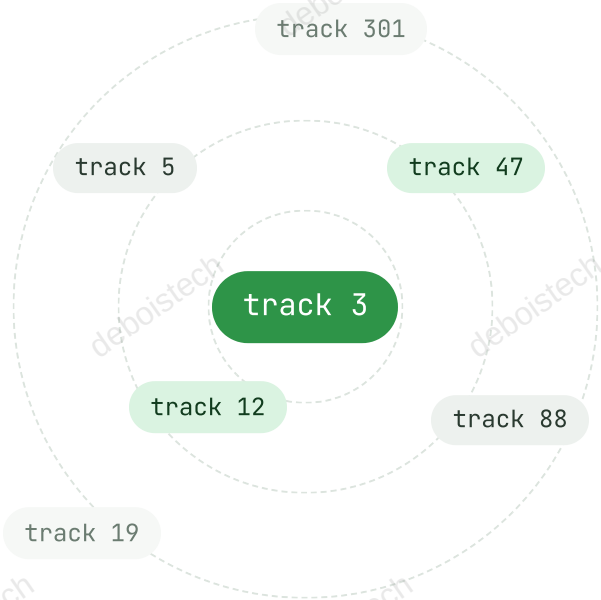
A SENTENCE

the | fluffy | **cat** | sat | down

A PLAYLIST

track 1 | track 2 | **track 3** | track 4 | track 5

Same window, same objective. The model never knew it was language.



APPEARS IN THE SAME PLAYLISTS

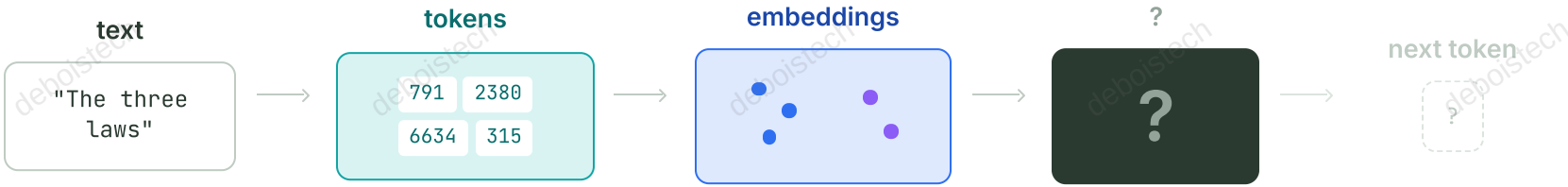
Songs get neighbours too

07 Wrap up

Three things to carry, and one box still shut.

SHORT RUN · HOW WE GOT HERE · TWO PHASES · LANDSCAPE · TOKENS · EMBEDDINGS

The spine, filled in

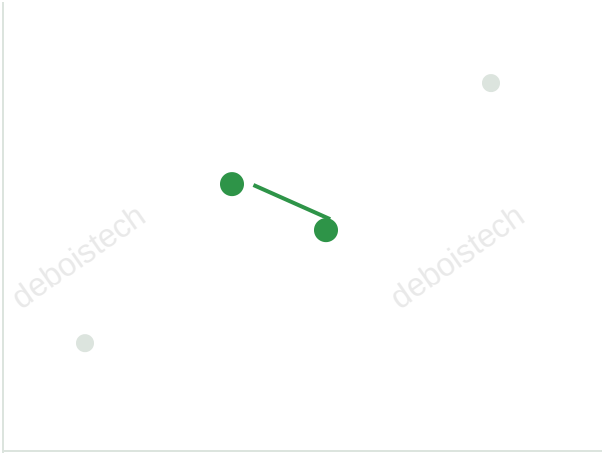




→ cost

→ context

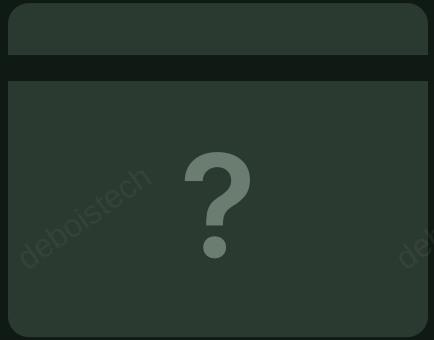
→ the weird bugs



similar means close.



you live here.



WHAT'S STILL INSIDE

attention

the next-token loop

why it's slow

Lecture 2 · inside the box

ATTENTION

THE NEXT-TOKEN LOOP

WHY IT'S SLOW

Bring the notebook you broke today. We start from your token counts.